

**Preliminary Report
And Design Document**

Team: PROMETHEUS

12.05.2024

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1 PROGRESS REPORT

1.1. PROGRESS STATEMENT

Frankly, things could definitely be better. Our team is attempting to cover the aftereffects of the dreaded ill fortune that Murphy's Law inevitably brings at any given point of a project's development - anything that can go wrong will go wrong. Aside from the fact that our old model was ineffectually modeled, due to the inexperience we had during the first stages with using 3d modelling software, we also had the wonderful (not really) experience of dealing with brittle materials and the disappointing reality that 3d printing materials weren't nearly as durable as we thought they were initially.

Ultimately we scrapped the "butterfly" prototype we initially used and wound up testing out an entirely different disk design - aside from the fact that we're now utilizing the space we have far more efficiently, we now have space for a GPS module as well - something we couldn't fit in our prior design.

1.2. TASKS LIST

1.2.1. WHAT'S DONE

- Purchase of all necessary supplies
- Completion of all internal project documentation for team coordination
- Testing and confirmation of proper operation of all electronic hardware components
- Completion of integration of all software components
- Testing and confirmation of proper operation of spring deployment system and fabrication of external tube for disk probe deployment
- Testing and confirmation of proper operation of deployment mechanism within disk probes

1.2.2. WHAT'S NOT (WE'RE WORKING ON IT)

- 3D CAD model for the individual disk probes
- Initial fabrication of individual disk probes
- Testing of individual disk probes
- Final assembly
- Using a proper table for this section (We ran out of time)

1.3. DETAILED PROJECT STATUS

The journey we took was initially very promising, until we ran into our first roadblock - we didn't have enough space to fit a proper GPS system. This meant throwing out the 3d modelling of a portion of space the probes spread over, as we had no way of giving a reference point between the probes - but without many other real alternatives at the time, we ignored it and forged on, passing into the second stage, where we were very abruptly dealt a terrible hand by Murphy's Law - the PLA plastics we were working with were brittle, thick, and even more terrible for space efficiency than we initially thought - to the point where not only did the probes break upon impact with the ground, damaging any test equipment we initially loaded within the test prototypes, but the battery refused to fit.

Hoping we wouldn't have to resort to bending lithium ion cells to fit the impossible angles of our butterfly probes (a do-it-yourself approach to bomb making in any context), our group split to cover

ground in order to solve this issue - one half continued tweaking the design of the individual probes, in hopes of making the butterfly design more space efficient, while the other half attempted to find metal fabricators in the country that somewhat fit our limited budget in order to fix the issue of fitting our electronics into the sub probes. Out of time, out of options, and running out of money, we eventually wound up scrapping the entire butterfly prototype we ran up until Stage 2 with, and at the risk of losing a significant margin of points, we wound up changing our design entirely. However this itself wound up presenting problems, as the coordinating professors we were working with, who supplied us with computers that ran SolidWorks, the CAD software we were most used to using, were all gone for Easter break - and although we attempted to use free alternatives such as OnShape, we were left without sufficient time to provide a functioning 3D model to present for the FRR. Below, in section 2.3.2, we have a selection of hand drawn prototypes, drawn with slide rulers and compasses. Not ideal, but hey - it worked for Goddard. It'll maybe work for us. Maybe.

On the electronics side of things, our prototyping was quite successful, albeit with two major hiccups - our first was the loss of an experimental test rig, consisting of an HC-12 and an Arduino, due to the blunt force impact from one of the butterfly models snapping apart upon collision with the earth. The other consisted of the very rude awakening (of which we had little idea until we started testing) that multiple probes required multiple receiving modules - prompting us to double (plus one) our number of Arduino modules, which added a significant cost to our already costly endeavor. Ultimately whether we manage to finish by Friday remains to be seen. We cross our fingers and hope for the best.

2 MISSION OVERVIEW

2.1. MISSION PURPOSE

The principal goal of our CanSat mission is to collect basic environmental data through a set of smaller individual probes in order to properly assess the status of the airspace the CanSat(s) travel through. Potential impact ranges from environmental studies to pollution statistics with possible use cases such as air traffic and control and meteorological updates for civilian and military applications.

2.2. MISSION OBJECTIVES

1. Our choice of secondary mission was an unusual one. Instead of attempting to fit more sensors into a slow descent vehicle, our team opted for a more unconventional approach through the use of three (3) slow descent sub-probes, launched from the main body of the principle CanSat with the minimum requirements to fulfill the principle objectives: air pressure, pollution, and temperature. The projected estimate is that we will be able to create a three dimensional graph of air vectors, pressure differences, temperature differences and pollution flow through the use of multiple probes, thereby collecting vectorial data respective to the required atmospheric data.
2. Successful CanSat launch and recovery would be characterized through the following criteria: successful launch, sub-probe deployment, successful real-time data transmission, and recovery of all components related to the CanSat project.
3. Our research expectations would be best represented by the varied sub-probes and their properties, as their expected performance may vary from their real-life performance, as well as the ability to graph pressure differences and temperatures as vectors, along with predicting pollution vectors. We hope to better our system through testing, as the ability to predict

the aforementioned vectors in a package as small as the CanSat would prove useful in many domains.

4. Our expected tests are to measure the efficiency with which a dispersed slow-descent probe can project and predict pollution changes and pressure differences over a wide area, as well as gauge the possible effectiveness of small individual probes packed together in a single launch system. Our expected results are the collection of data within a "cone" of space, collected and processed by the aforementioned sub-probes, giving a wide area analysis potential to anyone with access to a small rocket, and a marked increase of efficiency in probe generation per launch, giving way to a more cost effective method of collecting data. We intend to contrast our direct results with meteorological data published on behalf of the European Union's Meteorological Infrastructure System, or EMI, through the use of EUMETSAT, a coordinated effort among countries in the EU for meteorological survey and use.

2.3. GENERAL DESIGN OVERVIEW

2.3.1. MECHANICAL/STRUCTURAL DESIGN

1. Mechanical Design: The CanSat sub probe structures is made of lightweight biodegradable PLA plastic printed in parts through the use of a 3D printer, which provides strength and durability while keeping the overall weight of the CanSat at a minimum, as well as providing an eco-friendly, biodegradable structure, while the canister itself is constructed of a thin yet durable commercial grade aluminium. The structure of the three (3) individual sub-probes is designed to be completely contained at launch, insuring a water-tight and resistant quality to the components residing within that guarantees safe storage and flight integrity, minimizing any present risks of deployment failure.
2. Components: The major components of the CanSat can be divided between the CanSat main tube and the sub-probes.

The sub-probes, henceforth referred to as "disk probes", is composed of disk, with a cut out within to house a spring actuated wing that deploys itself in the event of it's exit from the main tube, with a rectangular sliver of plastic affixed to the innermost end of the wing relative to the disk, in order to prevent the wing from ejecting entirely. The electronics housed inside are neatly packed, centered around the motherboard, which is positioned directly above the wing, separated by a thin plastic border, and on either side lie the battery and sensors.

The main tube, while of less remarkable interest, maintains it's role of simplicity, being constructed of only a spring and a containing tube with a short time delay mechanism, ensuring the contraption remains packed together until it has fully exited the rocket payload bay, whence forth it then disperses the three sub probes into the atmosphere.
3. Placement: The main board is housed directly adjacent to the left of the spring designed for full deployment of the CanSat sub-probes, the battery and sensor array being mounted around the main board, while the communications and pollution sensors will be mounted below and above the main board respectively
4. Explanation: On the sub-probes, the main board, in our case, the Arduino, acts as the central controller for the CanSat, processing sensor data and controlling the transmitter. The temperature/pressure sensor measures the temperature and pressure of the immediate environment during descent, the pollution sensor compiles data on the quantities of ozone and CO2 gasses present, and the transmitter sends the respective data back to the ground station. The battery provides power to the CanSat during the flight, and the sub probes are stored insomuch as they fit into each other, creating an efficient compact design during the

storage process - once deployed, the natural aerodynamic properties of each individual sub probe will allow them to scatter randomly, creating a wide spread for information gathering. The three sub-probes fit together during initial launch and flight of the CanSat, ejecting shortly after it's apogee. After main canister ejection, within 5-6 seconds, the main canister in turn ejects all three sub-probes, which represent the actual CanSat models of use to us. No longer constrained by the walls of the container, once fully exited, the sub probes each individually deploy a "wing", specifically a slightly asymmetrical biconvex wing. The wing presents itself as the arrest mechanism whereby the CanSat slows it's decent, converting vertical momentum drawn by the gravitational constant into angular momentum. The CanSat is detailed further in Figures 1 through 3; all measurements may be found in the aforementioned diagrams provided.

2.3.2. TECHNICAL DESIGN SCHEMATICS

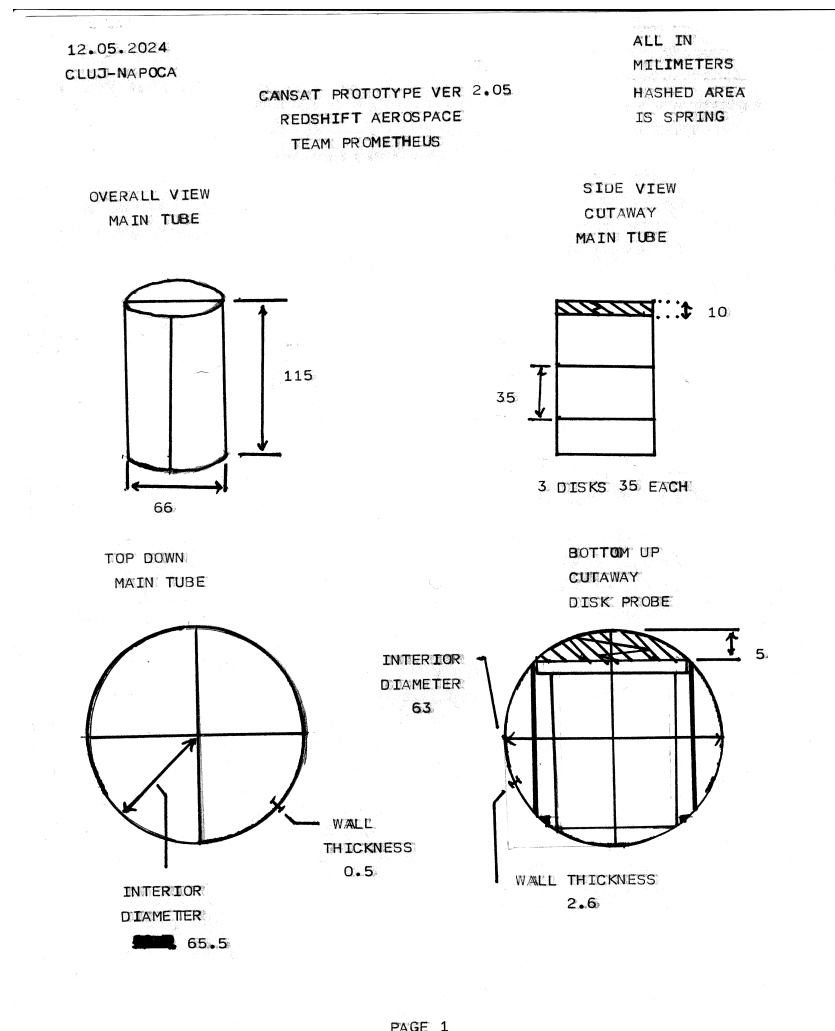
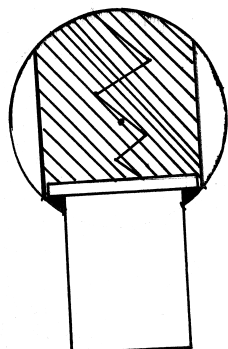


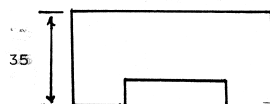
Figure 1:

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CLUJ-NAPOCA

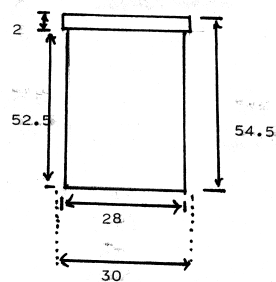
TOP DOWN
DISK PROBE
WING DEPLOYED



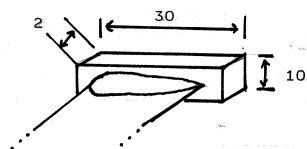
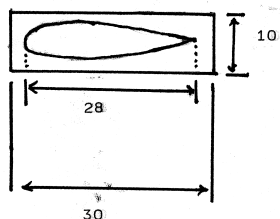
SIDE PROFILE
DISK PROBE



TOP DOWN
WING PROFILE



SIDE PROFILE
WING



3D MODEL
WING PROFILE

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Figure 2:

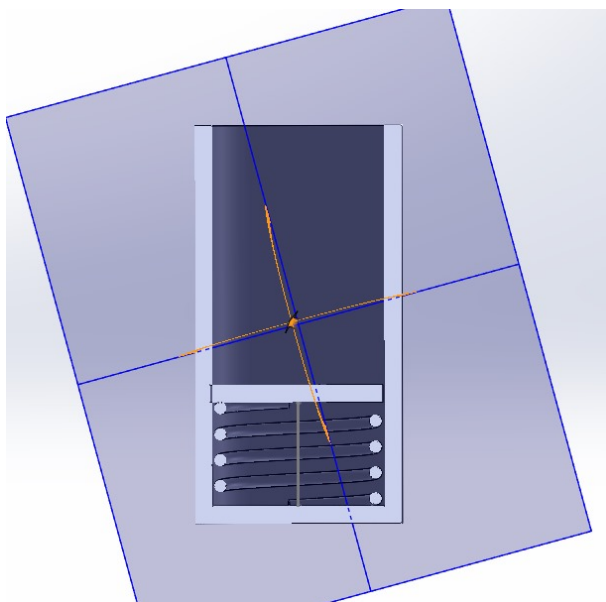


Figure 3:

2.3.3. ELECTRICAL DESIGN

1. **Electrical Interface:** An electronic drawing is provided in Fig 1.06, detailing the exact positioning of the respective components within the CanSat. The exact electrical components and their functions are as follows: the main circuit board is the Arduino mini, measuring at 17x33mm, the transmitter and receiver is an HC-12 measuring at 27.8x14.4mm, the battery is an unmarked 5v Li-On battery with a 1000mAh capacity measuring at 15x15mm, the gas pollution sensor is a MQ135, measuring at 20x32 mm, and the temperature/pressure sensor is a BMP280, measuring at 12.50x16.50mm. A GPS module can be found integrated through the NEO-6m, measuring at 25x35mm, with the antenna module measuring at 25x25mm. A detailed diagram of the electrical interface can be found in Figure 4.

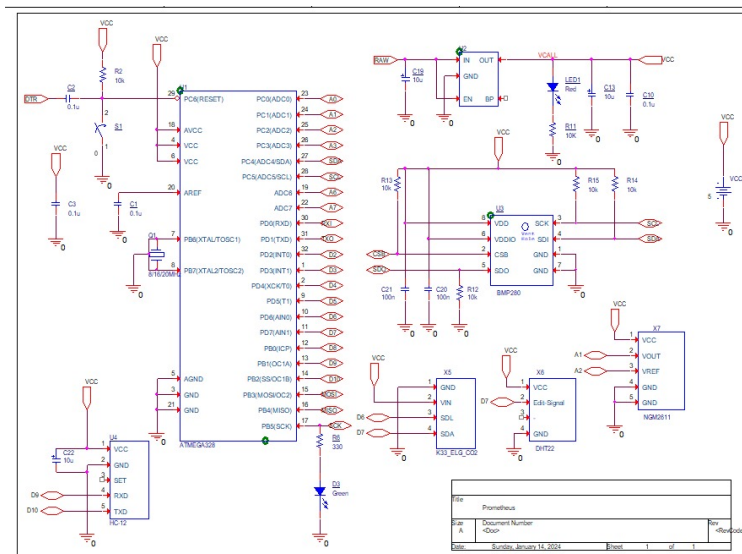


Figure 4:

2. **RF Link:** The CanSat uses an HC-12 RF transmitter for data communication with the ground station at distances of up to 1100 meters. The HC-12 is a transmitter that typically operates in the 433.4-473.0MHz frequency band - multiple channels can be set, with a stepping gap of 400 KHz, and there are in total 100 channels. The maximum transmitting power of module is 100mW (20dBm), the receiving sensitivity is -117dBm at the standard baud rate of 5,000bps in the air, and the communication distance is 1,000m in open space. Due to the presence of an MCU within the module, the user doesn't need to program the module separately, and the all transparent transmission mode is only responsible for receiving and sending serial port data, so it is remarkably convenient to use. The module adopts multiple serial port transparent transmission modes, and the user could select them by AT command according to use requirements FU1, FU2 and FU3 in idle state consuming 0.8mA, 3.6mA and 16mA respectively, while the maximum working current, that we will take advantage of, is 100mA. Below are two photos (Fig. 5,6) that represent the HC-12 integration in our project.



3. RF Link (Continued): As shown in the above figure, the HC-12 module is used in a physical wiring configuration when replacing half duplex communication. The left device sends serial port data to the module, and after RXD port of left module receives the serial port data, it will automatically transmit. The right module in turn automatically receives the data, and restores, from TXD, the serial port data originally sent by the left device. The process is identical from right to left. A limitation of the HC-12 is that while the half duplex state is available between modules, they unfortunately cannot receive and send data at the same time.

2.3.4. POWER BUDGET

- Arduino Mini: 4.5 mA average consumption
 - HC-12 Transmitter: 100mA peak consumption (Once every 5 seconds)
 - MQ135 Gas Sensor: 150mA peak consumption, 100mA average consumption
 - BMP280 Temperature and Pressure sensor: 2.74 μ A average consumption
 - NEO-6m GPS receiver: 45mA peak consumption (Once every 5 seconds)
 - Battery: 1.1 W
4. Power Consumption and Duration: The total power consumption of each subprobe of the CanSat is estimated to be 127.5 mA, while each individual battery has a capacity of 2000mAh, which should provide enough power for the CanSat to operate for a minimum of 4 hours, accounting for peak consumptions of the MQ135 and HC-12 modules.
 5. Battery: Each individual sub-probe uses one (1) 2000mAh Li-Ion battery (PKcell LP803860), rated for 3.7 volts

2.3.5. SOFTWARE DESIGN

1. Software Design: The CanSat's software is designed to control the various electronic components and manage the data collected by the sensors. The main board micro controller runs the software and controls the sensors, transmitter, and other components. The CanSat's software collects data gathered in real time from the aforementioned sensors specified in section 2.3 and transmits them every 5 seconds in regular intervals.
2. Data Gathering and Storage: The CanSat is expected to gather approximately 500 KB of data per flight distributed across the three probes. The data is stored on-board the CanSat unit through the Arduino Pro Mini unit, subsequently transmitted to the ground control unit through the HC-12 radio module.
3. Programming Language and Development Environment: The CanSat software is written primarily for the Arduino platform in Python with small segments written in C, and is developed using the Atom IDE development environment. We are entirely relying on the base Konsole command line that is found within the Kubuntu operating system, as it represents a simple base for interfacing with the HC-12 radio system.

2.3.6. RECOVERY SYSTEM

1. Description: Due to size limitations and constraints, our team has had no choice but to design the main body with no recovery system in mind - thankfully, the total weight of the main tube is under 30 grams, meaning that it will both naturally slow its descent, and will minimise any risk of causing injury by falling onto unknowing bystanders. Additionally, being made entirely of aluminum and steel, in the event it is lost, there is no real negative

financial or environmental effect. While a parachute might have been feasible under normal circumstances, our space budget means we cannot by any means fit a parachute into the probe, at least not with the amount of time or finances we had available.

Recovery of the disk probes will be ensured in double fashion both by the NEO-6m GPS module located within each CanSat disk probe, as well as the signal strength of each of the 3 HC-12 radio modules. The GPS module will read out its precise location once every 5 seconds, and if by some misfortune it malfunctions, the signal strength of the HC-12 module has been tested to be a reliable stand in for a location orientation device, presuming the seeker has sufficient time and patience to test by signal strength.

2.3.7. GROUND SUPPORT EQUIPMENT

1. Equipment: The ground support equipment for the CanSat mission consists of a 3 computers, 3 HC-12 antennas, and 3 arduino modules for each of the 3 sub-probes included in the design. The computers receive and compile the data received into a .txt file generated through the Konsole command line system on each computer, before being individually and manually compiled by the human operators of the three individual computers, rendering a visual model manually.
2. Transmitter Frequency: The transmitter frequency that is present within all CanSat sub-probes is between 433.4-473.0MHz. Through EU standard regulation, the aforementioned frequency range is free use - therefore, we will assign a specific frequency to each probe, presently assigned as 434.0, 435.0, and 436.0 mHz respectively, however we intend to remain flexible as unexpected radio interference is always a very common likelihood on the day of the launch.

3 TEST PLAN

3.1. TESTING PROCEDURES

1. Structural integrity and descent capabilities of the product are verified through a series of timed drops, in varied weather conditions, utilising the natural height of our universities' respective buildings (Approximately 20 meters in height) in order to fine-tune our initial descent rate eventually moving on to higher natural elements that exceed said buildings' height (such as cliff sides), thus achieving our desired rate of between 6 and 12 meters per second.
2. Basic data transmission, data acquisition, and processing testing, consisting of the assembly of the HC-12 on the Arduino Mini, along with all sensors, is verified through a series of distance tests, checking to see if the HC-12 can transmit a simple message initially over an increasing number of distances, after which, leaving entirely on battery power, we can commence quality analysis of the data from sensors and GPS modules by exposing them to the elements under different conditions and verifying if general meteorological readings generally align with expected values.
3. Tube deployment tests can be carried out simply by placing the disk probes within the main tube and repeating the drop tests enumerated during the structural integrity tests
4. Final assembly test can be performed by repeating the aforementioned drop tests, with the key difference being the disk probes and main tube containing the sensitive electronic components, demonstrating that the electronics will survive the eventual impact with earth.

3.2. TEST OBJECTIVES

1. Test Objectives: The overall test objectives are identical to the ones presented in the prior subsection:
 - (i) Verification of mechanical deployment of both main tube and disk probes, with integrated wing component, making sure each component deploys correctly under the conditions of stress imposed by launch and descent;
 - (ii) Verification of structural integrity by checking potential points of failure and establishing a general consensus of reliability of the structure housing outside of freak incidents;
 - (iii) Verification of hardware data transmission and sensor collection under conditions of stress and rapidly fluctuating temperature differences caused by sudden acceleration within the rocket;
 - (iv) Verification of serviceability of aforementioned electronic equipment upon impact with earth at the estimated descent rate of 8 meters per second.
2. Test Methodologies: Accurate consensus of data is acquired through the proper utilization of the basic scientific method: starting from a question, we experiment, analyzing data and creating a baseline metric upon which we might improve in future tests. Each test individually is carried out a minimum of four times to eliminate any drastic outliers that may be caused by unforeseen or unpredictable events such as catastrophic failure. We can split up the tests into two categories:
 - (i) Mechanical drop tests are carried out at the University of Physics Babes Bolyoi, in Cluj-Napoca, as the building has a consistent height of 20 meters from the third floor, providing a consistent factor in tests. Additionally, we used in our tests almost exclusively PLA 3d printed plastics and consumer grade aluminium for drop tests, supervised by professors who gave us informative tips on streamlining the process. Data was collected through two students utilizing stopwatches and a third filming on a budget high speed camera - allowing for post operation computer analysis - therefore we are able to accurately deduce the descent rate and the time after which the disk probe enters spin.
 - (ii) Electronic and software tests typically go hand in hand, as one verifies the other in turn. A large empty open field with a set of 20 measurements in 5 second intervals, followed by a 50 meter increase of distance was a sufficiently reliable test both for the transmission, base computation and sensor elements of the CanSat project.
3. Test Parameters: Criteria for drop tests largely depend on two parameters; opening of the canister within 6 seconds of initial release, and the stabilization of the disk probes within 20 meters of initial release. Electronic parameters centered around the estimated transmission distance of 1000 meters of the HC-12, along with the successful transmission of data verifying functionality of sensor and Arduino integration. Sensor parameters are the local meteorological readings received daily from sites such as Accuweather.
4. Safety Considerations: Safety considerations center mainly around the drop tests of the disk probes and main canister within the Babes-Bolyoi university, as blunt force impact from a densely loaded object given 20 meters to accelerate under earth's natural gravity poses a significant risk to one's health. Thankfully we were given the supervision and clearance to work with our professors at the department of physics, allowing us to use the only tall geographical landmark that was readily available to us for testing purposes. Of small note is the heated element within the 3d printer we utilized; throughout the printing process, we were isolated from the printing chamber by a 5mm sheet of glass, minimising any potential burn hazards.

5. Justification: The aforementioned tests are necessary in their entirety to fully grasp an understanding as to the reliability of the CanSat project conceived by our team as a whole. Drop tests are entirely necessary, both with and without electronic payload, in order to assure proper mechanical function, structural integrity, and electronic reliability under stress conditions, and electronic transmission tests not only verify the function of the Arduino base computer paired with the HC-12 radio module but also the sensor array attached to it, given we verify with local meteorological readings at time of measurement.

4 TEST RESULTS

4.1. SUMMARY OF TEST RESULTS

As earlier described, the test results can be divided into two categories: mechanical and electrical.

1. Electrically, the CanSat project is finalized and ready to deploy. The HC-12 integration with the sensor, GPS array and Arduino module, after initial hiccups with software integration, work flawlessly - a total of 20 tests were executed at ranges between 50 to 1000 meters in an open field, and the rate of failure of a data transmission packet was within acceptable levels for all three sub probes (under 5% failure rate), while sensors all reported similar if not identical readings to standard verified meteorological sources (Fluctuations $\pm 2^\circ$, CO2 and CH4 fluctuations within 50 ppm)
2. Mechanically however, the CanSat initial design was a total failure. Not only did the probe fail to stabilize within 20 meters for arrested descent, shock tests broke apart the probe upon impact, the interior probe size was far too small for our needs and, worst of all, we lost an Arduino + HC-12 combination in a loaded shock test. It was at this point we decided to abandon testing the butterfly design and move towards an alternative model - that may or may not work. As of right now, 12/05/2024, first test is scheduled on the 14th, with the 13th being sufficient time for 3d modelling. Unfortunately we haven o other real choice.

4.2. ANALYSIS OF TEST DATA. OBSERVATIONS:

While there isn't much of note to comment within the electronics side of the CanSat probes, as it was a simple matter of integrating readily available python code into a command line and testing it in a field over the course of a few hours, the mechanical side of the CanSat project yielded interesting, if not extremely disappointing results. The first round of tests of the "butterfly" test were entirely based on a faulty 3d model created by an inexperienced team member, which lead to us discarding the prototypes before we even flew them. The second and third batch of drop tests were rushed, as aside from needing to intersect with the fickle time tables of our university professors, the design we "borrowed" initially modelled after the American BLU-45 / Russian PFM-1 mine design, our original point of inspiration - but this model, at least when rendered physically through a 3D printer, tended to rip at a critical point of failure. Scrambling to fix the point of failure, it took another 3 batches of 4 "butterflies" for us to finally give up, when a total of 10 snapped upon impact out of 12, and only one ever began stabilizing after the initial 20 meter drop. While we understand the implications and possibility that the prototypes might have needed more height to stabilize, we cannot understate the impact of the wasted space and the toll it took on our design process - the final lost Arduino test was simply the final straw for our team before we discarded the idea entirely.

Below: The failed "butterfly" design"; initial design sometime October 2023.

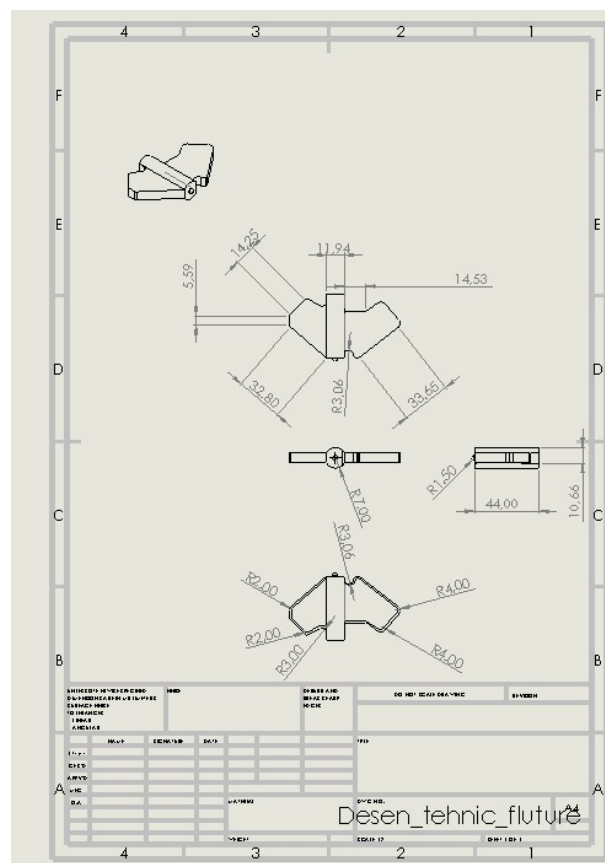


Figure 7:

5 CONCLUSION

5.1. SUMMARY OF THE FRR

In summary, we've gone a long way from the initial prototyping stage last year. While we are now presented with an incredible challenge - to manufacture and test an entirely new design within 5 days - we remain adamant that our disk probe design, which will allow the CanSat to fully utilize the space given to us while exploring alternative descent possibilities over a larger area of data collection, will be at least sufficiently interesting enough to change the predominant mentality of attacking the CanSat challenge - and hopefully, enough to grant us the win in the national stages.

5.2. OPEN POINTS

Our next steps include transporting the CanSat payload and it's eventual launch - and as we understand the possibility of failure points within transportation, we intend to be prepared in case of failure of any one module during transportation or misfire during launch. Due to budgetary constraints we are not able to create perfect clones of all three disk probes, however we have at least one backup planned for the launch day in case transportation or launch fails. We intend to test the disk probes until the morning of the 17th when we depart for Brasov, as long as last minute checks within the dorms of the city, in order to minimize failure to the greatest extent our human nature will allow us.